

ExecLab AI

Business case for an agentic execution-research platform

Thesis: portfolio alpha is not fully earned when an idea is approved; it is earned only after the position is implemented in the market. ExecLab AI helps portfolio managers, execution traders, and quant analysts protect alpha by turning pre-trade data, benchmark schedules, and execution cost logic into a defensible execution plan before the order hits the tape.

Pre-trade expected cost, market risk, and liquidity before sending	Scenario lab TWAP, VWAP, POV, IS, and hybrid schedule comparisons	Agent debate fast execution vs liquidity seeking recommendations	Custom algo chat English constraints become structured algo logic	Memo output cause-effect narrative with caveats and diagnostics
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The user: who specifically uses this

User	Concrete workflow	Why ExecLab AI matters
Portfolio manager / pod analyst	Before entering, exiting, or rebalancing a position, compare execution choices against expected alpha, liquidity, volatility, benchmark risk, and peer-name pressure.	Shows whether the idea survives implementation cost, then converts PM constraints into an executable plan without giving up edge.
Execution trader	Evaluate participation rate, benchmark risk, volume profile, adverse price movement, completion urgency, and natural-language constraints for a live or simulated order.	Turns intuition into an evidence-backed plan, custom schedule, and concise explanation for PMs, risk, or post-trade review.
Small fund / emerging manager	Replace broker-only post-trade reports, spreadsheets, and vendor dashboards with a lightweight pre-trade execution research layer.	Provides institutional-style pre-trade analytics and narrative decision support at a price small teams can justify.
Quant researcher / fintech product team	Prototype, test, and explain algorithmic execution logic using public market data, transparent schedule mechanics, and agent-generated custom algo plans.	Accelerates learning and product iteration without black-box vendor tooling.

What firms do today - and why it is still broken

Current approach	What happens at funds today	Gap ExecLab AI fixes
Broker fees / balance-sheet access	PMs and traders pay indirectly through commissions, flow, relationships, or balance-sheet economics, then receive broker TCA, algo color, or post-trade commentary.	ExecLab creates an independent pre-trade plan before broker selection: what to send, how fast, what benchmark to use, and why.
External TCA / analytics vendors	Asset managers buy services that surface cost, rank, and benchmark results, but expected-cost models often feel like another black box interpreted by another analyst.	ExecLab explains cause and effect: volume curve, urgency, participation, drift, beta/sector pressure, peer moves, and benchmark sensitivity.
Desk heuristics and default algos	Teams lean on standard practices: VWAP for benchmark safety, POV for liquidity, TWAP for simplicity, IS for urgency, plus rough max-participation rules.	ExecLab turns heuristics into scenario-tested recommendations and custom algo logic with a narrative that PMs, risk, and traders can defend.
Internal TCA hires / spreadsheets	Some shops hire execution specialists or build notebooks, but output depends on one person's reasoning, stale assumptions, and manual reconciliation across market, sector, peer, and order data.	ExecLab standardizes the workflow, reduces reasoning error, saves analyst time, and scales the same decision discipline across names.

The problem: alpha leakage without pre-positioning

Execution is a hidden P&L; line item. A PM can find a strong signal, size the position correctly, and still lose alpha through timing, impact, or benchmark slippage.

Most workflows are reactive. Broker post-trade TCA explains what happened after the order is done; ExecLab moves the story upstream into pre-trade positioning.

ExecLab explains cause and effect. It links strategy features - urgency, participation, concentration, and drift exposure - to direct cost, while separating own impact from market/sector and peer-name movement.

The business pain:

- PMs need to know whether execution cost overwhelms expected alpha before they trade.
- Traders need justified action: speed up, slow down, cap participation, split the order, or build a custom hybrid schedule.
- Custom algo requests often start as English PM constraints; ExecLab converts them into structured schedule logic.
- Leadership needs a narrative that connects strategy features to direct cost, market risk, and peer movement.

Business use case in one sentence: ExecLab AI helps a PM or trader ask, "Given this name, size, urgency, benchmark, peer context, and constraints, what is the cleanest way to execute without giving back the trade idea?" It saves vendor/headcount cost while producing explainable action, custom algo logic, and a cause-effect execution story.

Agentic AI solution: from market data to execution plan

The product does not simply summarize data. It decomposes the execution decision into specialist tasks, runs deterministic tools, compresses the evidence, and lets agents reason from the same grounded context.

Product workflow

Step	What happens	Business output
1. Fetch intraday data	Pull live or recent OHLCV bars and construct benchmark prices, volume shape, returns, and diagnostics.	A trader can see the day context before choosing a schedule.
2. Generate schedules	Create TWAP, VWAP, POV, implementation-shortfall style paths, plus user-defined hybrid schedules.	The system compares realistic alternatives rather than one default algo.
3. Simulate fills and TCA	Estimate arrival cost, VWAP cost, slippage, completion, participation, and schedule concentration.	Turns execution quality into measurable basis-point tradeoffs.
4. Decompose risk and cost	Use liquidity, volatility, size, volume-curve features, market/sector risk, and peer-name movement to separate own participation impact from systematic pressure.	Pre-trade recommendation is grounded in the name, the market tape, and comparable names.
5. Agent debate, custom algo, and memo	Debate agents recommend urgency vs liquidity, a custom algo agent converts English constraints into schedule logic, and narrative agents explain direct cost drivers.	PMs and traders get a defensible execution plan and memo in minutes.

Why this must be agentic

1. Execution decisions are conditional

The right answer changes with order size, urgency, liquidity, volume curve, volatility, and benchmark. A static rules engine cannot comfortably explain tradeoffs across all combinations.

2. PMs need reasoning, not raw metrics

A table can say VWAP beat TWAP by a few bps; an agent can explain whether that was volume shape, drift, participation pressure, beta/sector risk, peer movement, or schedule concentration.

3. Custom algos start as language

PMs and traders describe constraints like "do not exceed 12% participation" or "finish before the close if volatility spikes." Agent planning converts intent into structured parameters and custom schedule logic.

Deterministic tool layer

Data fetch, volume curves, schedule generation, fill simulation, TCA metrics, expected-cost features, beta/peer context, and chart-ready diagnostics. This layer keeps calculations reproducible and auditable.

Context engineering layer

Compresses raw bars and scenario outputs into structured facts: benchmark prices, liquidity profile, risk splits, caveats, and schedule-specific outcomes. Agents reason over evidence, not raw noise.

Agent layer

Specialist agents explain tabs, debate urgency vs liquidity, design custom schedule logic, write a final memo, and use a critic/golden-set review path to reduce hallucinated claims.

Core business capabilities

Pre-trade analytics: expected cost ranges, liquidity/volume profile, participation pressure, volatility context, market/sector risk, peer-name evaluation, and benchmark-risk preview.

Algo recommendation: recommends whether to prioritize speed, liquidity seeking, benchmark alignment, or a hybrid plan based on the trade context.

Custom algo design: transforms natural-language PM constraints into a structured execution plan that can be turned into schedule logic and later production algo templates.

Cause-effect TCA narrative: explains how strategy features such as urgency, participation cap, and schedule concentration translate into direct cost, market exposure, or peer-driven movement.

Outputs a PM or trader can use immediately:

- Scenario comparison table with cost and completion metrics.
- Analyst-style memo explaining the recommended execution approach and cost drivers.
- Debate view showing the tradeoff between urgency and liquidity.
- Market/sector/peer risk decomposition and caveats so the recommendation is safe, transparent, and auditable.

Economics: selling execution discipline as a scalable SaaS layer

ExecLab AI is an execution research and decision-support product - not a broker, OMS/EMS replacement, or guaranteed-alpha engine. The commercial value is alpha preservation, faster execution research, and cleaner communication between PMs, traders, and analysts.

Pricing strategy

Tier	Target buyer	Price	What is included	Why it works
Free / education	Students, demos, open-source funnel	\$0	Limited scenarios, public data, no saved workspace	Builds trust and showcases the execution simulator.
Analyst	Quant students, junior analysts, solo researchers	\$49 / user / month	More runs, saved scenarios, richer memos, exportable analysis	Low-friction self-serve product.
Pro Trader	Small fund trader, PM analyst, fintech product owner	\$299 / seat / month	Higher run limits, custom-algo planner, memo history, scenario templates	Cheaper than one bad execution decision and easy to expense.
Team Workspace	Emerging fund, student fund, fintech team	\$1,500 / month for 5 seats	Shared workspace, templates, team memo library, batch scenario analysis	Creates collaboration around execution research.
Institutional Pilot	Hedge fund pod, broker analytics team, OMS/EMS vendor	\$10k-\$25k / month pilot	Private deployment, data connector, compliance controls, custom metrics	Enterprise value comes from integration and workflow ownership.

Back-of-envelope one-user-month economics

Assumed active pro user: 20 trading days, 5 names/day, 4 schedule scenarios/name, and 60-120 agent memos/month.

Token usage: about 1.5M-4.0M input tokens and 0.2M-0.6M output tokens/month using compact structured context rather than raw bars.

Compute/data cost: Python calculations plus scale-to-zero Cloud Run style workloads. With public or licensed data, marginal cost is roughly \$10-\$45 per active user-month, depending on memo volume and model choice.

Illustrative margin:

- Analyst tier at \$49: viable for light users and funnel acquisition.
- Pro Trader at \$299 with \$25 COGS: about 90% gross margin before support and fixed costs.
- Team at \$1,500 with \$150-\$250 COGS: about 83%-90% gross margin.
- Break-even for a lean \$3k/month fixed cost base: roughly 11-15 pro seats or 2 team accounts.

Value justification for PMs and traders

Executed notional / month	Execution cost improvement	Estimated alpha preserved	Pricing implication
\$10M	5 bps	\$5,000 / month	A \$299 seat is easy to justify if it prevents one poor schedule choice.
\$50M	5 bps	\$25,000 / month	Team pricing becomes small relative to implementation shortfall savings.
\$100M	5 bps	\$50,000 / month	Institutional pilots can be justified by workflow, auditability, and integration value.

Why these technical choices support the business model

Technical choice	Business reason
Tool-grounded schedule and TCA calculations	Keeps recommendations explainable and auditable; users can see which metric drove the conclusion.
Context compression before LLM calls	Reduces token cost and improves reliability by sending concise evidence rather than raw intraday bars.
Specialist agents and handoff	Mirrors a real desk workflow: market-data review, pre-trade analysis, opposing views, custom plan, critic, final memo.

Trust posture: the current version is a bar-based execution research simulator. It should not claim queue position, venue routing, hidden liquidity, tick-level NBBO, or production fill accuracy. The enterprise path is integration with licensed tick data, broker execution logs, and internal TCA benchmarks.

Live demo: <https://execlab-ai-q7smatnpa-uc.a.run.app>